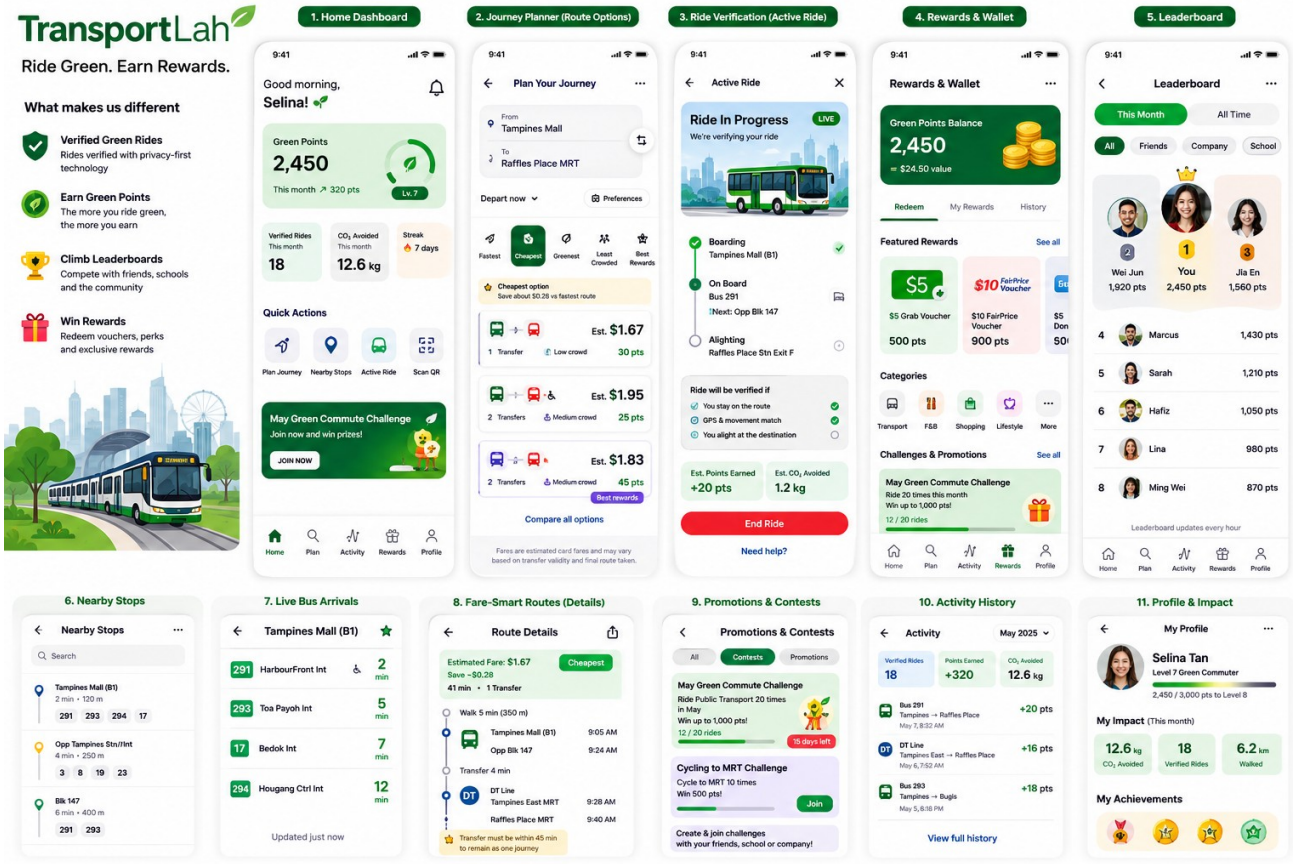
TransportLah Final Design Proposal

Ride green. Earn rewards. Climb the leaderboard.

A gamified, privacy-first green commute rewards platform for Singapore



Product thesis

Singapore already has strong journey planners and bus timing apps. TransportLah differentiates itself by turning verified public transport usage into Green Points, contests, public leaderboards, sponsor promotions and measurable CO2 impact.

Prepared: June 2026

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1. Executive Summary

TransportLah is proposed as a Singapore public transport companion app that rewards greener commuting. It combines live public transport data, lightweight ride verification, fare-smart route comparison, Green Points, public leaderboards, admin-managed promotions and sponsor-backed contests.

The app should avoid competing head-on with established transport utilities on journey planning alone. Instead, it should use transport data as the foundation and build a differentiated engagement layer: verified rides + rewards + sustainability + contests.

Recommended positioning
Singapore's green commute rewards and challenge platform, powered by public transport data and privacy-first ride verification.

Strategic Pillar	What It Means
Daily utility	Live arrivals, route options, train alerts, crowd context and fare-smart comparisons.
Trustworthy rewards	A lightweight confidence-scoring service validates rides before points and leaderboard credit are awarded.
Green identity	Users build a visible profile around verified rides, CO2 avoided, streaks and achievements.
Campaign platform	Admins and sponsors can create contests, promotions and public or organisation-specific leaderboards.
Privacy-first design	Raw GPS is used only during active ride verification and retained temporarily.

2. Product Vision and Target Users

2.1 Product Vision

TransportLah should make public transport usage rewarding, social and measurable. The product experience should be built around a simple loop: Plan ride -> Verify ride -> Earn points -> Join contests -> Climb leaderboard -> Redeem rewards.

The primary tagline is: Ride green. Earn rewards. Climb the leaderboard.

2.2 Target Users

Segment	Needs	TransportLah Value
Daily commuters	Reliable commute information and motivation to use public transport.	Live arrivals, fare-smart routes, verified points and rewards.
Students	Affordable travel and social challenges.	School leaderboards, badges and low-cost commute nudges.
Office workers	Daily commute planning and employer sustainability initiatives.	Company challenges, off-peak incentives and team dashboards.
Families	Simplicity, savings and reward visibility.	Profile impact summaries, wallet and family-friendly challenges.
Sponsors	Measurable green campaign participation.	Admin promotions, contest registration and reward fulfilment.
Schools / companies	Behaviour-change campaigns.	Verified team leaderboards and aggregate impact reporting.

3. Market Research and Competitive Positioning

Singapore already has a mature transport app ecosystem. Existing apps are strong in live bus timing, multimodal route planning, train alerts, fare-card account services and taxi/private-hire booking. TransportLah should therefore not be positioned as another generic journey planner.

3.1 Competitor Review

App	Current Strengths	Opportunity for TransportLah
MyTransport.SG	Official LTA app with personalisation, nearby transport information, MRT/road alerts and multimodal journey planning.	No strong public green rewards, contest registration, sponsor promotion or verified leaderboard layer.
SimplyGo	Travel expenditure/history, card balance, card services, top-ups, reward points and journey planning.	Strong payment and card utility, but not a broad public green commute challenge platform.
Citymapper	Best-in-class multimodal navigation, live departures, disruptions, step-by-step route guidance and saved stops/lines.	Excellent navigation, but not locally positioned around green rewards, verified commuter identity or public contests.
Moovit	Public transport planning, live arrivals, maps, service alerts and step-by-step directions.	Utility-focused; weak on local sponsor campaigns, verified rewards and leaderboards.
SG BusLeh	Bus-first UX, bus location on map, capacity bars, unified search and EZ-Link balance support.	Strong bus timing utility, but not a full green mobility engagement platform.
SBS Transit app	Real-time bus arrivals, bus route details, delayed/diverted services, train delays and disruption alerts.	Operator-centric; no broad multi-operator reward and contest layer.
Grab / CDG Zig	Taxi/private-hire booking, payments, promotions and trip history.	Competes for mobility wallet attention but does not primarily reward public transport use.

3.2 Strategic Insight

- Basic bus arrivals, nearby stops, route planning, MRT alerts and favourites are table stakes.
- TransportLah should focus investment on features that existing transport apps do not strongly cover: verified green rides, Green Points, public leaderboards, contests, admin promotions and sustainability reporting.
- The user promise should be: Save money, ride greener, earn rewards.

Strategic conclusion

Use LTA transport data as the foundation, but build the differentiated value layer above it: verified rides + Green Points + contests + leaderboards + promotions + sustainability reporting.

4. LTA DataMall API Foundation

The LTA DataMall guide provides a practical foundation for live public-transport and mobility features. Relevant APIs include Bus Arrival, Bus Services, Bus Routes, Bus Stops, Passenger Volume, Train Service Alerts, Facilities Maintenance, Station Crowd Density Real Time and Forecast, Planned Bus Routes and Bicycle Parking.

Important design constraint: passenger-volume APIs are aggregated and generated for previous-month data. They help with analytics and planning, but they do not prove that an individual user took a specific ride. Therefore, TransportLah needs its own lightweight ride verification layer unless a future authorised fare-card integration is secured.

Feature Area	Relevant LTA APIs	Use in TransportLah
Live bus arrivals	Bus Arrival	ETA, current estimated bus location, load/crowding, wheelchair accessibility and vehicle type.
Bus network context	Bus Services, Bus Routes, Bus Stops	Service validation, stop geofencing, route matching and first/last bus rules.
Train context	Train Service Alerts, Station Crowd Density Real Time/Forecast, Facilities Maintenance	Disruptions, crowd levels and accessibility alerts.
Analytics	Passenger Volume by Bus/Train and OD datasets	Demand patterns, campaign insights and aggregate planning. Not user ride proof.
Active mobility	Bicycle Parking	Bike-to-transit challenges and station-adjacent

		mobility nudges.
Road/mobility context	Traffic Incidents, Speed Bands, Estimated Travel Times	Phase 2 multimodal context and comparison to car travel.

4.1 API Integration Principles

- Do not call LTA APIs directly from the mobile client. Use a backend proxy to protect the AccountKey.
- Cache high-volume data such as bus stops, routes and services.
- Use active-stop polling for Bus Arrival rather than constantly polling every stop.
- Handle 500-record pagination via $skip where applicable.
- Treat LTA data as transport context and validation support, not as user ride evidence.

5. Core User Features

5.1 Journey Planner

Users can search by origin and destination and compare route options. The planner should support bus, MRT/LRT and mixed-mode journeys, but the MVP should not try to outbuild Citymapper or Google Maps. Its priority is to support fare-smart, green and reward-aware route decisions.

5.2 Live Bus Stop Screen

The live bus stop screen shows services arriving soon, ETA, destination, stops away, load/crowding, wheelchair accessibility, vehicle type and whether arrival time is monitored from bus location or schedule-based.

Bus Load	UI Treatment
SEA - Seats Available	Green badge / low crowd indicator
SDA - Standing Available	Amber badge / medium crowd indicator
LSD - Limited Standing	Red badge / high crowd indicator

5.3 Train Status Screen

The train module shows train service status, affected lines/stations, free public bus or MRT shuttle notices during disruptions, real-time and forecasted station crowd levels, and lift maintenance notices.

5.4 Green Ride Check-in

The check-in experience should be voluntary and transparent. The app should say “Start Green Ride” and “We are verifying your ride,” rather than implying surveillance or perfect certainty.

6. Lightweight Ride Verification Service

TransportLah should use a confidence-based ride verification engine instead of continuous tracking or a single GPS signal. The service should classify rides as verified, probable, low-confidence or rejected based on a combination of stop proximity, service validity, bus arrival context, route matching, speed plausibility and alighting location.

6.1 Verification Flow

1. User taps Start Ride.
2. App captures 1-3 GPS points near a stop or station.
3. Backend checks proximity to a valid bus stop or train station.
4. User confirms service or route option.
5. Backend validates that the service serves the selected stop and that the stop is in the right sequence.
6. Backend checks live or recent Bus Arrival context where relevant.

7. During the ride, the app samples GPS lightly and preferably calculates first-pass route matching on device.
8. User taps End Ride, or the app suggests auto-ending near a downstream stop/station.
9. Backend computes verification score and awards points according to verification tier.
10. Raw GPS trace is deleted or downsampled after the retention window.

6.2 GPS Sampling Strategy

Ride Phase	Sampling
Start check-in	1-3 points
First 3 minutes	Every 10-15 seconds
During ride	Every 30-60 seconds
Near expected alighting	Every 10-15 seconds
After ride ends	Stop completely

6.3 Ride Confidence Score

Signal	Purpose	Weight
Starts near valid bus stop	Plausible boarding point	15
Selected service serves stop	Route validity	10
Bus Arrival context matches check-in time	Bus plausibly arrived	20
Movement follows bus route corridor	Strong movement validation	25
Speed pattern is bus-like	Rejects walking/cycling/implausible travel	10
Remains near route corridor	Prevents drive-by check-ins	10
Exits near downstream stop	Plausible alighting point	10

Score	Decision	Reward Impact
80-100	Verified ride	Full points and leaderboard eligible
60-79	Probable ride	Partial points; not high-value contest eligible
40-59	Low confidence	Personal stats only
Below 40	Rejected or flagged	No points

6.4 Route Matching Logic

```
ride_is_plausible =
  user starts near stop N
  AND selected service serves stop N
  AND user moves in the direction of stop N+1, N+2, N+3...
  AND user remains within 100-200m of the route corridor
  AND speed is within a bus-like range
  AND user exits near a downstream stop
```

6.5 Verification Tiers

Tier	Verification	Reward Impact
Tier 0	Manual ride claim	Personal stats only
Tier 1	GPS near valid stop + plausible route	Basic points
Tier 2	GPS + LTA live bus context + route following	Full points
Tier 3	Tier 2 + QR/NFC/device integrity	Contest/reward eligible
Tier 4	Future fare-card/SimplyGo partner verification	Highest trust

7. Fare-Smart Routes

Fare-Smart Routes helps users compare routes by estimated fare, travel time, crowd level, Green Points and CO2 avoided. In Singapore, public bus and MRT/LRT fares are distance-based, so the cheapest route is usually the lowest valid chargeable journey distance, subject to transfer rules.

7.1 Route Comparison Experience

Route Option	Time	Est. Fare	Transfers	Crowd	Green Points
Fastest	32 min	$1.95	2	Medium	25
Cheapest	41 min	$1.67	1	Low	30
Least crowded	45 min	$1.83	2	Low	35
Best rewards	39 min	$1.83	2	Medium	45

The route card should explain trade-offs in plain language, e.g. “Save about $0.28, but the journey may take 9 minutes longer.”

7.2 Fare Estimation Logic

- Total route distance and route sequence.
- Bus route stop distance and station origin/destination.
- Transfer limits and timing windows.
- Same-service and same-station re-entry constraints.
- Fare profile: adult, student, senior, concession or tourist.
- Monthly pass status, where relevant.
- Clear labelling as estimated unless official fare-source integration is used.

7.3 Transfer Validity Warnings

Situation	App Warning
Long transfer wait	This route may be charged as a new journey if the transfer wait is too long.
Same bus service repeated	Taking the same bus service again may be treated as a new trip.
Same MRT station re-entry	Exiting and re-entering the same train station may break fare continuity.
Too many transfers	This route may exceed the maximum transfer limit.
Very long route	This may be treated as a separate journey depending on total duration.

Fare-Smart positioning
Save money, ride greener, earn rewards.

8. Gamification: Green Points, Badges and Levels

The Green Points engine should reward consistent public transport usage, not only distance. It should encourage verified bus/train rides, multimodal journeys, off-peak travel, low-crowd choices and challenge participation.

Activity	Points
Verified bus ride	10
Verified train ride	12
Bus + train multimodal journey	25
Off-peak ride	+5
High-confidence verification	+5
Weekly streak: 5 public transport days	+50

Monthly green commuter badge	+100
Bicycle parking discovery/check-in near station	+5

8.1 Badges

Badge	Criteria
First Green Ride	First verified ride
Bus Hero	20 bus rides in a month
Rail Champion	20 train rides in a month
Multimodal Master	10 mixed bus/train journeys
Off-Peak Optimiser	10 off-peak rides
Carbon Saver	Estimated 25 kg CO2 avoided
Consistency Champion	4-week public transport streak
Verified Commuter	10 high-confidence rides

Levels: Seedling -> Sprout -> Leaf -> Tree -> Rainforest -> Green Champion.

9. Promotions, Contests and Leaderboards

9.1 Promotions Section

Admins can publish campaigns, sponsor offers and reward announcements. Promotions should be discoverable from the home dashboard and rewards tab.

Promotion Field	Description
Promotion title	Short campaign name
Sponsor	Brand or organisation
Description	Campaign details
Start / end date	Validity period
Eligibility	Public, school, company or location-based
Reward type	Voucher, points, badge or lucky draw
Verification requirement	Manual, GPS, high-confidence or QR/NFC
Status	Draft, scheduled, live or expired

9.2 Contest Registration

Users should be able to register for contests from the app. Each contest defines transport modes, scoring rules, required verification tier, leaderboard type, prize pool, privacy mode and anti-fraud rules.

Contest	Scoring
Monthly Green Commuter	Total verified public transport rides
Bus Champion	Verified bus rides only
MRT Explorer	Unique MRT/LRT stations visited
Off-Peak Saver	Off-peak rides
Team Green Challenge	Company/school team points
Neighbourhood Challenge	Area-based leaderboard
High-Confidence Hero	Only Tier 2+ rides count

9.3 Public Leaderboards

Leaderboards should support overall, weekly, friends, teams, neighbourhood, mode-specific and verified-only views. Public rankings should use display names and should count verified rides only.

10. Admin Portal

The admin portal turns TransportLah into a campaign platform. It should allow authorised admins, sponsors, schools and companies to create promotions, configure contests, review suspicious rides and manage rewards.

Admin Role	Capabilities
Super Admin	Full access
Campaign Manager	Promotions and contests
Sponsor Admin	Sponsor-specific promotions
Support Agent	User queries and reward fulfilment
Fraud Reviewer	Review suspicious ride claims
Analyst	Read-only dashboards

- Create, schedule and retire promotions.
- Configure contest rules, verification tiers, reward pool and leaderboard visibility.
- Review suspicious ride claims and top contest winners before reward fulfilment.
- Export contest results and aggregate impact reports.
- Publish push notifications and in-app announcements.

11. System Architecture

11.1 Mobile App

Recommended stack: Flutter or React Native for iOS and Android. Main modules: Home, Plan, Live Arrivals, Active Ride, Rewards, Contests, Leaderboards, Profile, Admin web views and Notification Centre.

11.2 Backend Services

Service	Responsibility
User Service	Login, profile, privacy preferences and display name.
Journey Service	Route search, live arrivals and station status.
LTA Data Service	API integration, caching, pagination and AccountKey protection.
Lightweight Ride Verification Service	GPS/event-based ride confidence scoring.
Fare Optimisation Service	Fare estimation, transfer validity and route comparison.
Gamification Service	Points, badges, levels and streaks.
Contest Service	Registration, scoring, eligibility and contest lifecycle.
Leaderboard Service	Public/private rankings and Redis sorted sets.
Promotion Service	Admin-created offers and campaigns.
Reward Service	Voucher issuance and redemption status.
Fraud Service	Anomaly detection and manual review.
Notification Service	Push, email and in-app alerts.

11.3 Data Layer

Component	Recommended Technology
Transactional data	PostgreSQL
Geospatial queries	PostGIS
Active ride state	Redis
Leaderboards	Redis sorted sets
Event processing	SQS, Kafka or Kinesis
Promotion images	S3/object storage
Analytics	BigQuery, Redshift, Snowflake or Databricks

11.4 Reference Flow

```
User searches journey
  -> Journey Service generates route candidates
```

```
-> LTA Data Service enriches with arrivals/stops/routes
-> Fare Optimisation Service estimates fare and transfer validity
-> Gamification Service estimates points
-> App displays Fastest, Cheapest, Greenest and Best Rewards options
```

```
User starts ride
-> Ride Verification Service validates stop/service/context
-> Redis stores active ride state
-> GPS/sensor proof packet is processed
-> Verification score is computed
-> Points ledger is updated
-> Leaderboard is refreshed
```

12. Data Model

Entity	Key Fields
User	user_id, display_name, email, privacy_mode
Ride	ride_id, user_id, mode, service_no, start_stop, end_stop, start_time, end_time, distance_km, verification_status
RideVerification	ride_id, verification_score, verification_tier, route_match_score, gps_quality_score, speed_score, lta_context_score
GreenPointsLedger	ledger_id, user_id, ride_id, points, reason, timestamp
Badge	badge_id, name, criteria, icon
UserBadge	user_id, badge_id, earned_at
Contest	contest_id, title, rules, start_date, end_date, status, required_verification_tier
ContestRegistration	contest_id, user_id, registered_at
LeaderboardEntry	contest_id, user_id, points, rank, rides, co2_saved
Promotion	promo_id, title, sponsor, description, image_url, start_date, end_date
Reward	reward_id, user_id, promo_id, voucher_code, redemption_status
FraudSignal	signal_id, user_id, ride_id, reason, severity, status
FareEstimate	route_id, fare_profile, estimated_fare, transfer_validity_status, warnings

13. Fraud, Privacy and Consent

13.1 Fraud Controls

- Location plausibility and stop geofence checks.
- Bus service and route validation.
- Speed plausibility and impossible movement detection.
- Duplicate ride and repeated short ride detection.
- Daily point caps and minimum ride distance/duration.
- GPS spoofing and device integrity checks.
- QR/NFC verification for high-value contests.
- Manual review of top contest winners and suspicious rides.
- Verified-only leaderboards for public contests.

13.2 Privacy Model

Data	Retention
Raw GPS trace	24-72 hours
Flagged ride trace	Until review is complete
Aggregated ride summary	Long term

Leaderboard score	Contest duration + audit window
CO2 and points summary	Long term

The app should use display names on public leaderboards, let users hide exact ride history, and explain that raw location data is used only during active ride verification. Long-term records should store summaries, not precise movement traces.

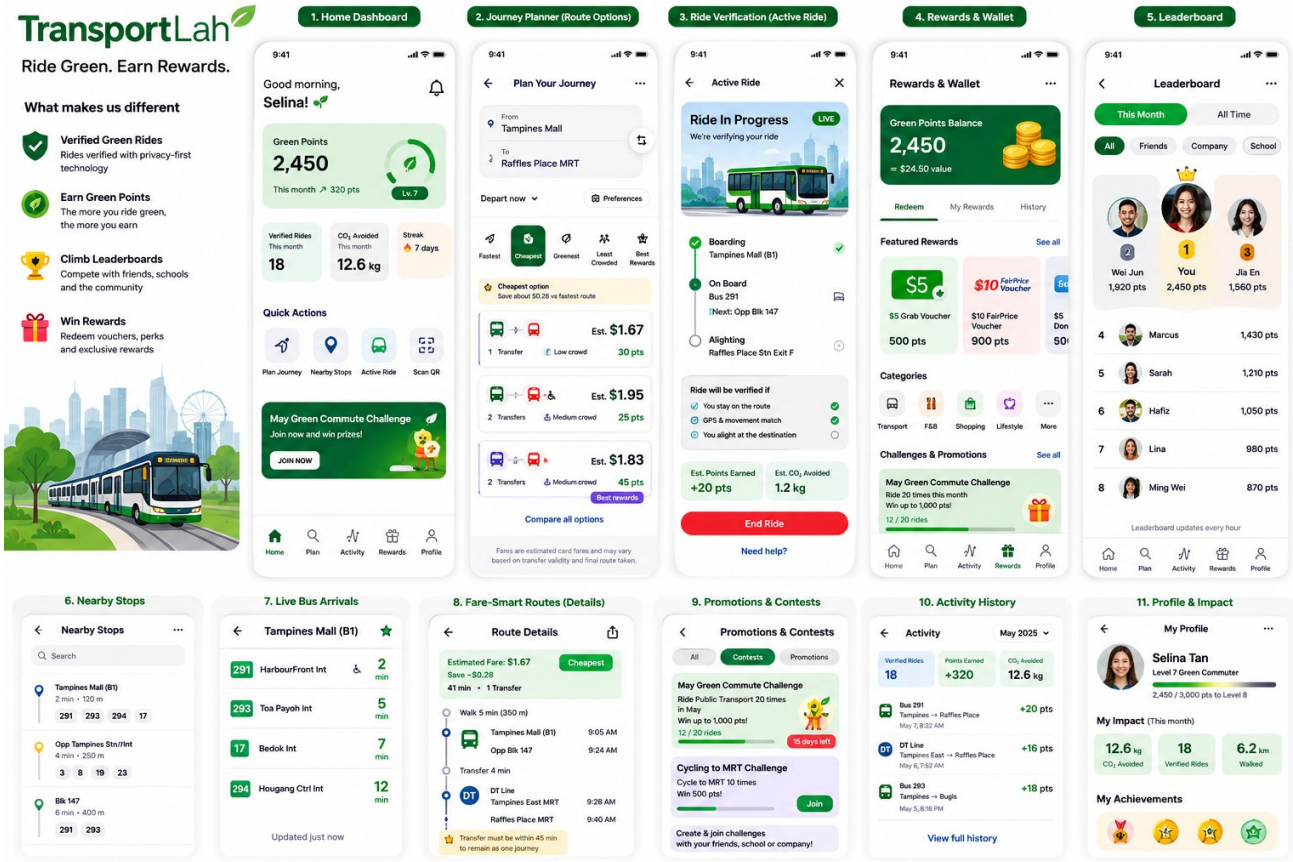
14. Proposed Wireframes and App Experience

The proposed UX follows a reward-led flow: Plan ride -> Verify ride -> Earn points -> Join contests -> Climb leaderboard -> Redeem rewards. The screens are designed to feel like a daily green mobility companion rather than a surveillance app.

14.1 Key Screens

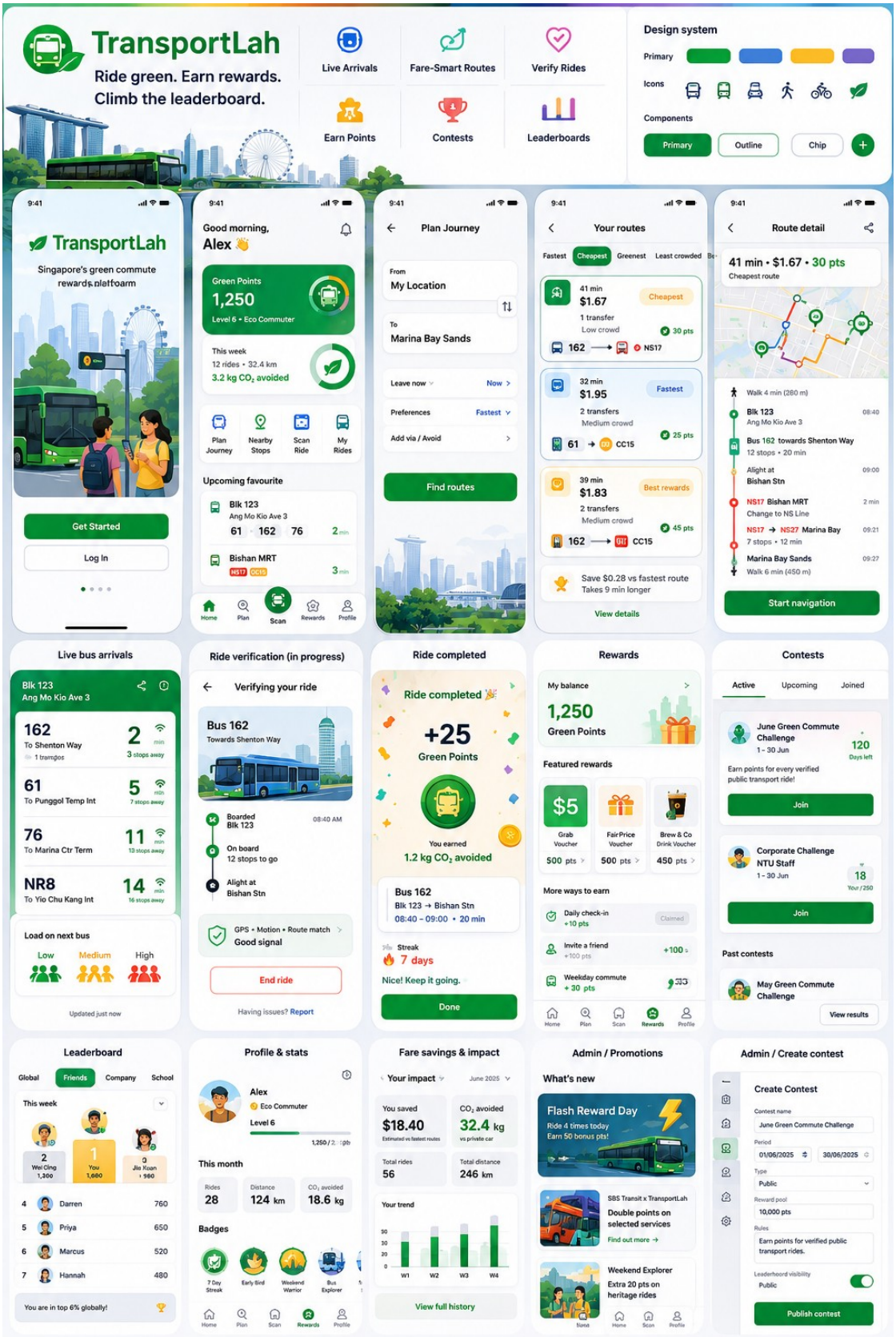
Screen	Purpose
Landing / Onboarding	Explain “Ride green. Earn rewards. Climb the leaderboard.”
Home Dashboard	Show Green Points, verified rides, CO2 avoided, streak and quick actions.
Journey Planner	Search routes and choose fastest, cheapest, greenest or best rewards.
Route Options	Compare time, fare, crowding, transfers and points.
Route Detail	Show step-by-step journey and Start Ride Verification CTA.
Live Bus Arrivals	Show ETA, load/crowd, destination and live validation context.
Ride Verification	Show ride progress and verification checklist.
Ride Completed	Celebrate points and CO2 avoided.
Rewards Wallet	Show points, featured rewards, categories and history.
Promotions and Contests	Discover, register and track campaign progress.
Leaderboard	Show global/friends/company/school ranks.
Profile and Impact	Show user level, monthly impact, badges and privacy settings.
Admin Promotions	Manage banners, sponsor content and engagement metrics.
Admin Create Contest	Configure contest dates, rules, rewards and verification tier.

14.2 Wireframe Board 1



Wireframe board 1: Home dashboard, route options, active ride verification, rewards, leaderboard and supporting screens.

14.3 Wireframe Board 2



Wireframe board 2: Detailed app screens, design system, admin screens and full product flow.

14.4 UX Language Principles

Use	Avoid
Start Green Ride	Start tracking
We are verifying your ride	We are monitoring you
Ride verified with high confidence	We know you were on the bus
Raw location is used only during ride verification	Always-on GPS tracking

15. MVP Scope and Roadmap

15.1 MVP Must-Haves

Priority	Feature	Reason
1	Lightweight ride verification	Core differentiator and reward integrity.
2	Green Points wallet	Creates reward loop and user identity.
3	Public verified-only leaderboard	Creates social competition.
4	Admin promotions	Enables campaign monetisation.
5	Contest registration	Converts rewards into structured engagement.
6	Fare-Smart Routes - basic	Adds everyday utility and savings angle.
7	CO2 avoided estimate	Supports green positioning.
8	Live bus arrivals	Table-stakes daily utility.
9	Train alerts and station crowd info	Table-stakes daily utility.
10	Basic fraud controls	Protects leaderboards and rewards.

15.2 MVP Exclusions

- Full fare-card integration.
- Payment wallet or card top-ups.
- Real voucher marketplace at launch.
- Best-in-class turn-by-turn journey navigation.
- Continuous GPS streaming or long-term raw GPS storage.
- Advanced ML recommendations.

15.3 Phase 2 Enhancements

- SimplyGo or fare-card partner integration.
- Corporate green commute dashboards.
- School/team challenges and verified commute certificates.
- Sponsor reward marketplace.
- AI commute coach and personalised green nudges.
- Advanced fare optimiser with fare profiles and pass logic.
- Predictive crowd avoidance and multimodal active-mobility rewards.
- QR/NFC verification for high-value campaigns.
- ML-based fraud detection.

16. Success Metrics

Category	Metrics
Product	DAU, WAU, ride check-ins per user, verified ride completion rate, retention.
Engagement	Contest registrations, leaderboard participation, reward redemption, notification CTR.
Sustainability	Verified public transport rides, estimated CO2 avoided, off-peak share, multimodal journeys.
Commercial	Sponsor campaigns, promotion views, contest joins, reward fulfilment

	and repeat sponsors.
Operational	LTA API cache hit rate, API error rate, verification latency, fraud review queue, leaderboard refresh latency.

17. References

- LTA DataMall API User Guide & Documentation, Version 6.8, 21 Apr 2026: Uploaded source document
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- PTC public transport fares and passes: https://www.ptc.gov.sg/fares/public-transport-fares-and-passes/

Note: Competitor research is based on public app listings and official pages available during proposal preparation. Feature availability may change over time.